

AI Developers Course — Final Project Presentation

HR Copilot

An AI-based Personal Assistant for Recruiters

Scenario #3 — screening candidates & automating scheduling

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Build: HRCopilot_Project_v2 — n8n-orchestrated

LangGraph · 6 nodes

RAG · Qdrant Cloud

n8n · 27-node workflow

Flask backend

Docker Compose

3 guardrails

THE PROBLEM

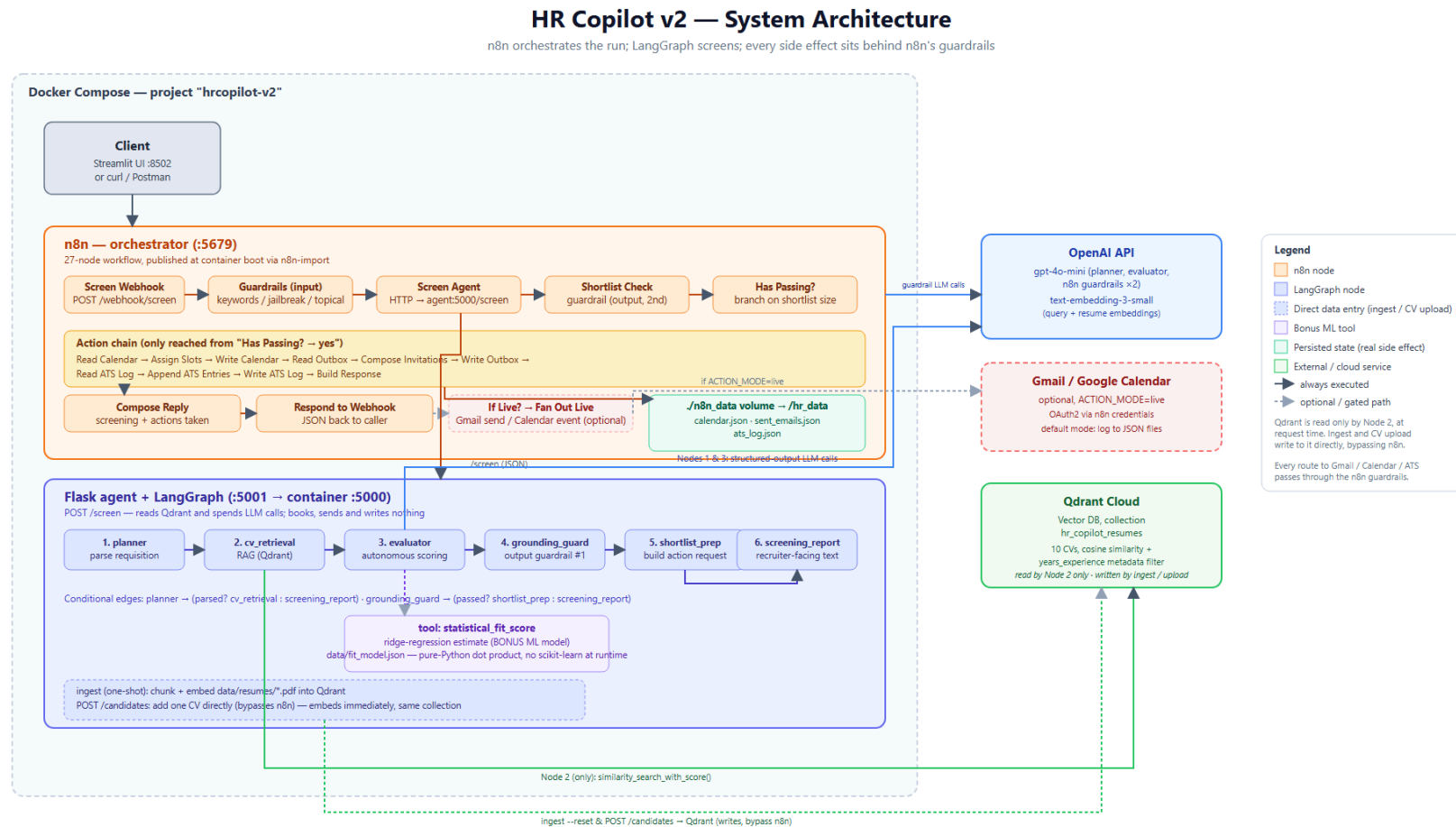
Screening is slow, manual, and easy to get wrong

- Recruiter reads a stack of resumes against one requisition
- Decides who is worth an interview — **inconsistent between recruiters**
- Manually finds a slot, writes an invite, updates the ATS
- Strong candidates get skimmed too fast; interviews double-book

What HR Copilot does instead: free-text requisition in → retrieved & scored candidates → booked interview, sent invite, logged ATS row — in about a minute, with two independent safety nets.

ARCHITECTURE

n8n orchestrates. LangGraph screens. Nothing bypasses the guardrails.



The graph screens; it never acts

1 **Planner** — parses free text into structured requirements

2 **CV Retrieval (RAG)** — similarity + metadata filter on Qdrant

3 **Evaluator (autonomous)** — scores each CV, calls one tool

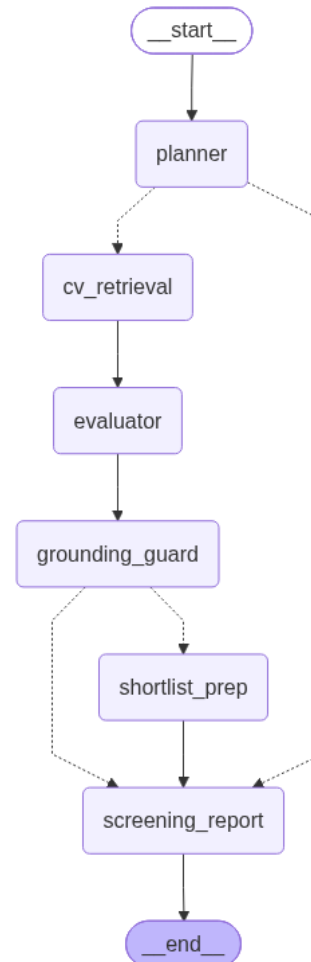
4 **Grounding Guard** — output guardrail #1, deterministic

5 **Shortlist Prep** — builds n8n's action request, no LLM

6 **Screening Report** — recruiter-facing summary

Two conditional edges branch on real LLM outcomes — not parsed → skip to report; nobody passed → skip the action request entirely.

Compiled graph visualization



More than plain similarity search

Piece	Choice
Source data	10 synthetic CVs (PDF), 8 professions, AI-generated per spec \$7
Chunking	<code>RecursiveCharacterTextSplitter</code> , 1200 chars / 200 overlap — every chunk carries the full CV text , never a fragment
Embedding	<code>text-embedding-3-small</code> , 1536-dim, cosine — Qdrant Cloud
Advanced retrieval	Metadata filter on years of experience, at query time — with graceful fallback if it excludes everyone
Query construction	Embed <code>title + required_skills</code> , not the raw requisition — matches how CVs are phrased

Every route to a real side effect passes through all of them

Guardrail	Where	Catches	On a hit
Input (keywords / jailbreak / topical)	n8n, before the agent runs	off-topic input, prompt injection	agent never called — zero LLM cost
Grounding Guard	LangGraph Node 4, deterministic	a claimed skill the CV doesn't evidence	justification rewritten before it can leave
Shortlist relevance	n8n, LLM-based	a shortlist from the wrong profession	blocks the run — nothing booked

The redesign story: both output guardrails used to run in the *last* node — after interviews were booked and mail was sent. They could describe a bad outcome, never prevent one. Moving them earlier is the main reason the architecture changed.

One webhook drives the whole run

Screen Webhook → Guardrails → Screen Agent (HTTP → LangGraph) → Shortlist Check → Has Passing? → Calendar / Mail / ATS → Respond

- 27-node workflow, published **before n8n's process starts** (n8n only registers webhooks at boot)
- Action chain runs **once per shortlist**, not per candidate — avoids a real read-before-write race across candidates' bookings
- `ACTION_MODE=log` (default): writes calendar/email/ATS JSON — a real persisted state change, zero Google setup
- `ACTION_MODE=live` : real Gmail + Google Calendar, fired after the webhook already responded

PACKAGING

One command, zero manual steps

- One `Dockerfile`, reused by 3 services (agent / ui / ingest)
- 5 Compose services: `n8n-import`, `n8n`, `ingest`, `agent`, `ui`
- Qdrant runs as a hosted Cloud cluster, not a local container

```
cp .env.example .env
# paste OPENAI_API_KEY, QDRANT_URL, QDRANT_API_KEY

docker compose up --build -d
```

Credentials + workflow import/publish happen **before** n8n's own process starts — no manual n8n login, no manual webhook activation.

7 scenarios cover the full decision surface

#	Case	Outcome
1	Senior Backend Engineer	clean match — books, invites, logs
2	Data Analyst	multi-candidate, disagreement flag on a boundary case
3–4	Off-topic / prompt injection	blocked in n8n, agent never called
5	Ship captain (zero-match)	legit input, honestly rejected on merit
6–7	SRE / ML Engineer (unseen by the fit model)	LLM judgement still holds out-of-sample

BONUS — spec §4.6, optional, not yet a final commitment

A statistical model as a second opinion

- **Ridge regression**, 5 features, 294 hand-labelled pairs
- Served as a **pure-Python dot product** — scikit-learn never ships in the runtime image
- Bound as a tool in Node 3 — the model must reason past it, not repeat it

MAE 0.77 pts, 89% band accuracy (grouped cross-validation). Barely beats the single best feature — its real value is the disagreement signal, not raw accuracy.

Honest framing: the core system (Sections 1–8) is complete and correct with or without this tool — the Evaluator degrades gracefully to CV-only reasoning without it. It recovers half the strong candidates; the misses are the genuinely ambiguous pairs, and closing that gap needs more training data rather than a different model.

Thank you

HR Copilot — questions welcome